

Mental Health and Covid-19

"How heavily did a state's COVID-19 measurement policy have an impact on mental health compared to its counterparts?"

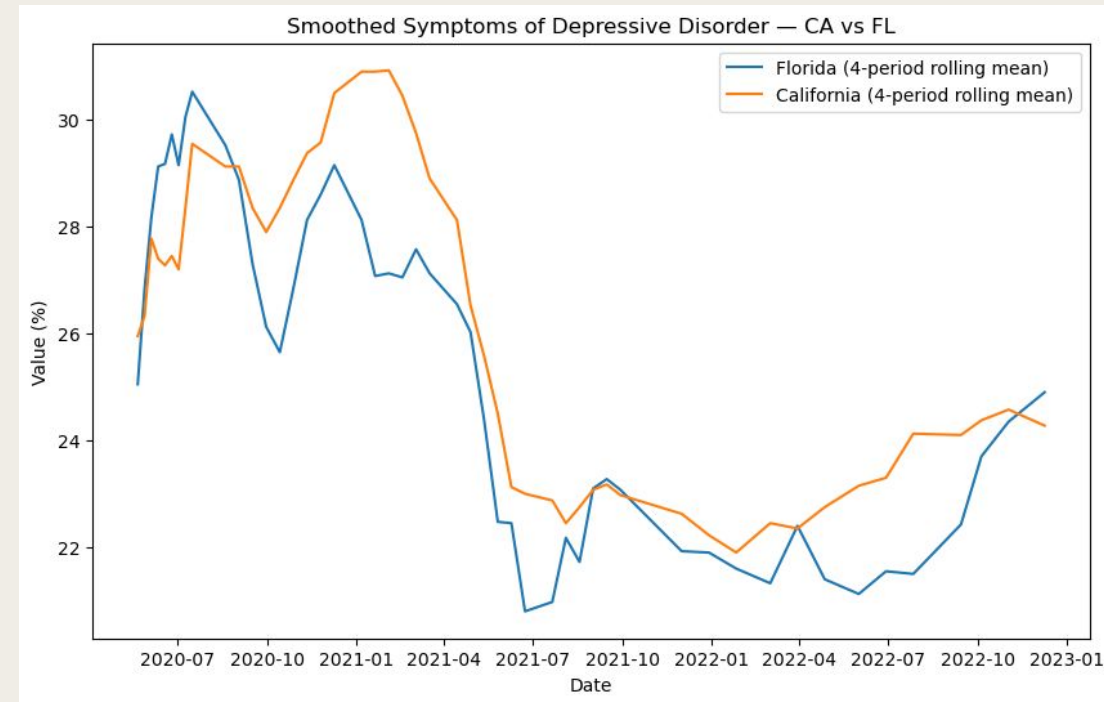
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Data

Dataset	Source	Shape	Time Frame
Data Source 1 Indicators of Anxiety/ Depression Rate	Data.gov	16,794 x 14	April 2020 - December 2024
Data Source 2 Covid-19 Government Response Tracker	Oxford Study (GitHub)	390,909 x 50	January 2020 - December 2022
Final Dataset		11,412 x 50	April 2020 - December 2022

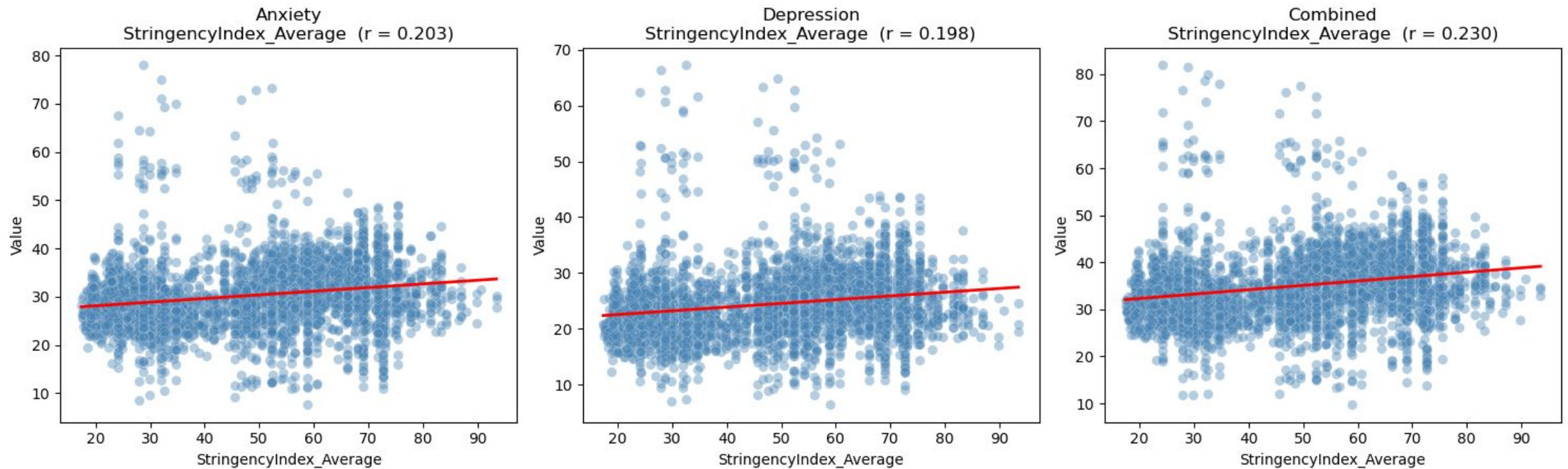
Target: Value

Variables: StringencyIndex_Average, EconomicSupportIndex



EDA Findings

StringencyIndex vs. Value by Indicator



Baseline Model vs. Simple Linear Model

Constant Model RMSE = 7.51

Simple Linear Model RMSE = 7.31

Improvement: 0.2pts (~2.72%)

$$\text{Value} = 30.47 + 0.093 * \text{StringencyIndex}$$

A real, but small improvement: Our feature variable cannot predict the depression/anxiety rate by its own

Did It Improve?

Constant baseline: test RMSE = 7.27

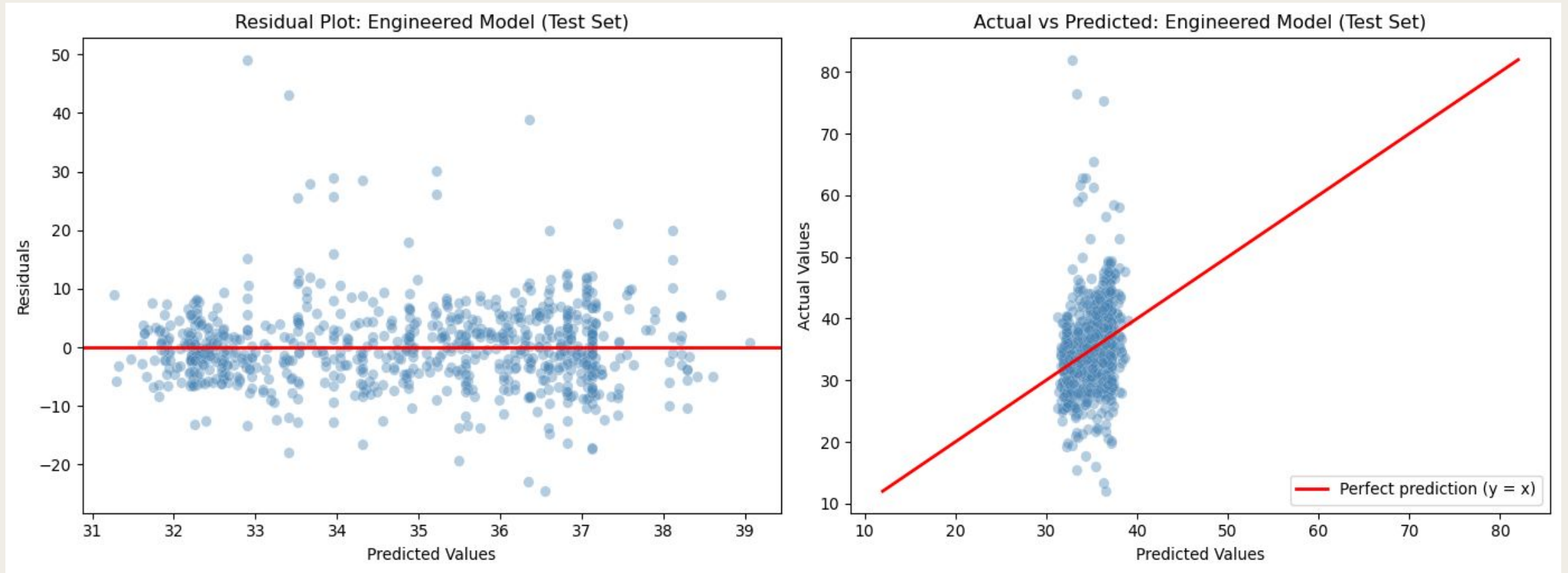
Phase 3 SLR (retrained): test RMSE = 7.07 → +2.72% improvement over baseline

Engineered Feature Model: test RMSE = 6.99 → +3.88% improvement over baseline

Best model beats the baseline by only 0.27 points of symptom prevalence — a bit better than the SLR, but still a rather weak to moderate gain

Model	MSE (train)	RMSE (train)	RMSE (test)	Improvement over Baseline (test RMSE)
Constant Model	57.2857	7.5687	7.2722	N/A
Phase 3 SLR (retrained)	54.2612	7.3662	7.0747	+0.1975 (+2.72%)
Engineered Feature Model	53.6210	7.3226	6.9900	+0.2821 (+3.88%)

What It Still Gets Wrong.



The model's predictions are squeezed into a narrow 33–39 band while real Values range from roughly 10–80, meaning stringency alone cannot capture the state-level demographic, economic, and temporal context that actually drives symptom rates.

What You Would Do Next

- **Additional feature:** Add EconomicSupportIndex as a primary predictor. Phase 3b showed well-behaved magnitudes and financial relief likely cushioned mental health impact independent of stringency
- **Modeling technique:** Gradient-boosted tree to capture non-linear relationships and feature interactions stringency alone cannot